Notes on the continuum sigma limit

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1 Introduction

2 Green's function transformation

To start our analysis, consider a linear differential operator L acting on $\mathbb{R}^n$, and its Green's function $G(x, y)$, defined by

$$LG(x, y) = \delta(x - y). \tag{1}$$

Introduce a variable s, and define a heat kernel $K(s, x, y)$ by the equation

$$\frac{\partial}{\partial s}K(s, x, y) = LK(s, x, y), \tag{2}$$

subject to the initial conditions $K(0, x, y) = \delta(x - y)$. If the operator L is dissipative in the sense that $\lim_{s\to\infty} K(s, x, y) = 0$, we can readily prove the relation

$$G(x, y) = -\int_0^\infty K(s, x, y)ds. \tag{3}$$

To prove this relation, act with L on both sides, to find

$$LG(x, y) = -\int_0^\infty \frac{\partial}{\partial s}K(s, x, y)ds = -K(s, x, y)\Big|_0^\infty = \delta(x - y), \tag{4}$$

which proves the identity assuming that L can be interchanged with the integration. With these definitions, one can define a generalized Weierstrass transform W_s through the convolution

$$W_s f(x) = \int K(s, x, y)f(y)d^n y, \tag{5}$$

which also satisfies the heat equation

$$\frac{\partial}{\partial s}W_s f(x) = LW_s f(x), \tag{6}$$

subject to the initial conditions $W_0 f(x) = f(x)$. This leads to the relation

$$f(x) = -L \int_0^\infty W_s f(x) ds, \tag{7}$$

which is a way of writing the solution to the potentential equation $Lu(x) = f(x)$.

As an interesting special case, consider the operator L to be the Laplacian Δ acting on $\mathbb{R}^3$, for which the heat kernel is given by

$$K(s, x, y) = (4\pi s)^{-3/2} e^{-|x-y|^2/4s}. \tag{8}$$

In this case, equation (3) becomes

$$-\frac{1}{4\pi|x-y|} = -\int_0^\infty (4\pi s)^{-3/2} e^{-|x-y|^2/4s} ds, \tag{9}$$

which can be readily checked with the change of variables $s^2 = |x-y|^2/4s$. The transform W_s is then the generalized Weierstrass transform in three dimensions. If the function $f(x)$ is a charge density, denoted $\rho(x)$, then we find

$$V(x) = \frac{1}{\varepsilon_0} \int_0^\infty W_s \rho(x) ds, \tag{10}$$

where $V(x)$ is a solution of Poisson's equation

$$\Delta V(x) = -\frac{\rho(x)}{\varepsilon_0}, \tag{11}$$

and ε_0 is the vacuum permittivity.

3 Application to quantum mechanics

The operator identity proved above is very general, as it can be applied to any dissipative operator L. The identity can also be extended to operators that are not dissipative in a strict sense, such as the Euclidean Dirac operator $\not{\partial}$, which will be demonstrated in a separate publication. Here we will consider the physically interesting case of the Hamiltonian operator $\hat{H}$ of quantum mechanics, and use the resulting identity to develop an approximate approach to finding the ground state of a quantum many body system. To start, we consider the energy expectation value of a quantum state $|\psi\rangle$

$$\langle \hat{H} \rangle_\psi = \langle \psi | \hat{H} | \psi \rangle . \tag{12}$$

The well known variational principle states that this energy expectation value should always exceed the ground state energy of the system for a normalized wavefunction $|\psi\rangle$. To apply the identity used above, we define the state $|\rho\rangle$ by the equation

$$\hat{H} |\psi\rangle = |\rho\rangle . \tag{13}$$

This is the potential equation $Lu = -f$ for the operator $L = -\hat{H}$, and therefore we can write the solution $|\psi\rangle$ in the form

$$|\psi\rangle = \int_0^\infty |\rho(\tau)\rangle \, d\tau, \tag{14}$$

where $|\rho(\tau)\rangle$ is the solution to the imaginary time Schrodinger equation

$$\frac{\partial}{\partial \tau} |\rho(\tau)\rangle = -\hat{H} |\rho(\tau)\rangle \tag{15}$$

subject to the initial condition $|\rho(0)\rangle = |\rho\rangle$. We note that this equation is always dissipative when $\hat{H}$ has a positive spectrum, and in the case where the ground state energy is not positive, we can consider a shifted expectation value

$$\langle \hat{H} \rangle_{\psi, E_s} = \langle \psi | \, (\hat{H} + E_s) \, |\psi\rangle - E_s, \tag{16}$$

and do the same analysis for the shifted potential $\hat{H} + E_s$, where $E_s + E_0 > 0$. For ease of notation, we will consider the case where the spectrum of $\hat{H}$ is positive.

Using the above equations, we can rewrite the energy expectation value $\langle \hat{H} \rangle_\psi$ in the form

$$\langle \hat{H} \rangle_\psi = \int_0^\infty \langle \rho(\tau) \, | \, \hat{\rho}(0) \rangle \, d\tau. \tag{17}$$

To make this equation look more symmetric, we note that the solution to equation (15) is $|\rho(\tau)\rangle = e^{-\tau \hat{H}} |\rho\rangle$, which allows us to write

$$\langle \hat{H} \rangle_\psi = \int_0^\infty \langle \rho | \, e^{-\frac{1}{2}\tau \hat{H}} e^{-\frac{1}{2}\tau \hat{H}} \, |\rho\rangle \, d\tau = 2 \int_0^\infty \langle \rho(\tau) | \rho(\tau) \rangle \, d\tau. \tag{18}$$

We also have to consider the normalization condition $\langle \psi \, | \, \psi \rangle = 1$, which can be written

$$\langle \psi \, | \, \psi \rangle = \int_0^\infty \langle \rho(\tau_1) | \rho(\tau_2) \rangle \, d\tau_1 d\tau_2 = 4 \int_0^\infty \langle \rho(\tau) | \rho(\tau) \rangle \, \tau d\tau, \tag{19}$$

where we have made the change of variables $\tau_\pm = (\tau_1 \pm \tau_2)/2$ and integrated over τ_-.

Let us consider minimizing the expectation value $\langle \hat{H} \rangle_\psi$ subject to the normalization condition. Using a Lagrange multiplier, this amounts to extremizing the functional

$$\mathcal{L}(\lambda, \psi) = \langle \hat{H} \rangle_\psi - \lambda(\langle \psi | \psi \rangle - 1). \tag{20}$$

However, after using the identities above, an additional set of constraints must be placed on the function $|\rho(\tau)\rangle$, namely that it satisfies equation (15). In order to impose these constraints we must introduce an infinite number of Lagrange multipliers at each timestep. To this end, introduce a basis for the Hilbert space $|\chi_\alpha\rangle$, which can be infinite-dimensional. Define an

infinite number of Lagrange multipliers $v_\alpha(\tau)$, and then define the sum $|v(\tau)\rangle = \sum_\alpha v_\alpha(\tau)\,|\chi_\alpha\rangle$. To impose the constraints, we will extremize the functional

$$\mathcal{L}(\psi, \lambda, v(\tau)) = \langle \hat{H} \rangle_\psi - \lambda(\langle\psi|\psi\rangle - 1) + \int_0^\infty d\tau\, \langle\rho(\tau)|\, (\overleftarrow{\partial_\tau} + \hat{H})\, |v(\tau)\rangle\,. \tag{21}$$

Using equations (18) and (19), then extremizing, we find

$$\int_0^\infty d\tau\, \langle\delta\rho(\tau)|\, \Big((2{-}4\lambda\tau)\,|\rho(\tau)\rangle + (-\overrightarrow{\partial_\tau} + \hat{H})\,|v(\tau)\rangle \Big) = \langle\delta\rho(0)|v(0)\rangle - \langle\delta\rho(\infty)|v(\infty)\rangle\,. \tag{22}$$

Because this is true for all variations, we immediately find boundary conditions for the function $|v(\tau)\rangle$, namely

$$|v(0)\rangle = |v(\infty)\rangle = 0. \tag{23}$$

Therefore, we find that an extremum must satisfy

$$(2 - 4\lambda\tau)\,|\rho(\tau)\rangle = (\partial_\tau - \hat{H})\,|v(\tau)\rangle\,, \tag{24}$$

for some function $|v(\tau)\rangle$ satisfying Dirichlet boundary conditions. Operating on the left with $\langle\rho(\tau)|$, we also find

$$\frac{\partial}{\partial\tau}\,\langle\rho(\tau)|v(\tau)\rangle = (2 - 4\lambda\tau)\,\langle\rho(\tau)|\rho(\tau)\rangle\,, \tag{25}$$

Note that given an eigenfunction $|E\rangle$ of the Hamiltonian, the solution to these equations is $|v(\tau)\rangle = 2E\tau e^{-E\tau}\,|E\rangle$ with $\lambda = E$ and $|\rho(\tau)\rangle = Ee^{-E\tau}\,|E\rangle$.

We have therefore found that the eigenfunctions of the Hamiltonian should correspond to solutions of equation (22) subject to Dirichlet boundary conditions. These results are exact, and therefore are equally difficult to solve as the original minimization problem. However, these calculations lead to a natural set of approximations to find the ground state energy. To see how this works, we will make the "simplest" approximation to the function $|\rho(\tau)\rangle$, namely that it is a Slater determinant[1]. We will define approximate dynamics using a time dependent Hartree-Fock approximation, and then minimize the energy subject to the normalization constraint.

Before proceeding with the mathematical details of this approach, we will make a few general remarks.

- The Slater determinant approximation is not the only approach to the problem. Any ansatz for the wavefunction $|\rho(\tau)\rangle$ can be used, with a corresponding approximation to the dynamics of the imaginary time Schrodinger equation.
- The Slater determinant ansatz is not equivalent to the usual Hartree-Fock method. To see this, note that equation (14) is an integral over $|\rho(\tau)\rangle$, so that even though $|\rho(\tau)\rangle$ is assumed to be a Slater determinant for given τ, the wavefunction $|\psi\rangle$ is not a Slater determinant. Therefore, we expect that this approximation will include correlation in the wavefunction.

[1] In this paper we will consider Fermionic systems, but the method can be extended to systems with Bosonic degrees of freedom.

- We expect this method to be best for including dynamic correlation, as opposed to static correlation. This is because the integral in equation (14) naturally includes different length scales in the problem, as evidenced from the intuition gained from the case of the three dimensional heat kernel (8). For static correlation, one must include many contributions from a single length scale, which is not naturally included in this integral. However, this approach could likely be combined with other methods for including static correlation in the wavefunction.

- Even though a variational optimization will be applied to approximate the ground state energy, there is no gaurantee that this energy will be higher than the true ground state. This is because the approximation to the time dependence does not necessarily retain the property that the ground state energy is always lower than the integral appearing in equation (18).

With these general ideas in mind, we will proceed with the analysis of the Hartree-Fock type approximation to this problem.

4 Slater determinant integral approximation

References

[1] B. Hall. *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*. Graduate Texts in Mathematics. Springer, 2003.

[2] Michael G. Eastwood and John Ryan. Aspects of dirac operators in analysis. *Milan Journal of Mathematics*, 75(1):91–116, Dec 2007.